# Devixo Solutions - AI/ML Internship

## Task 02: Build and Compare Machine Learning Models Report

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**Internship Track:** Artificial Intelligence & Machine Learning

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**Status:** Successfully Completed & Verified

## 1. Executive Summary

This report documents the rigorous development, evaluation, and comparative analysis of multiple machine learning models engineered to solve a real-world classification problem. Fulfilling the benchmarks set by Devixo Solutions, this project explores the complete machine learning lifecycle, emphasizing the critical impact of rigorous data preprocessing, robust feature engineering, systematic evaluation metrics, and hyperparameter optimization.

## 2. Dataset & Data Preprocessing

To ensure statistical validity, a real-world dataset comprising over 1,000 multi-attribute records was utilized. The raw data underwent a strict preprocessing pipeline prior to model training:

- **Missing Value Treatment:** Handled effectively using median imputation strategies to prevent skewness.
- **Duplicate Removal:** Cleaned dataset instances to eliminate redundant records and prevent training bias.
- **Categorical Encoding:** Converted qualitative categorical variables into quantitative formats using Label Encoding.
- **Feature Scaling:** Standardized numerical variables via scikit-learn's `StandardScaler` to establish zero mean and unit variance.
- **Dataset Partitioning:** Divided data into training (80%) and testing (20%) subsets for unbiased validation.

## 3. Model Development & Evaluation

Three distinct machine learning paradigms were trained and tested to evaluate comparative predictive performance: **Logistic Regression** (baseline linear model), **K-Nearest Neighbors** (non-parametric distance classifier), and **Random Forest Classifier** (ensemble decision tree method). Performance was assessed using accuracy, precision, recall, and F1 score.

## 4. Comparative Performance Analysis

| Model Name | Training Time | Primary Advantages | Key Limitations |
|---|---|---|---|
| **Logistic Regression** | ~0.012 s | Highly interpretable, extremely fast computation speed. | Assumes linear separability; underperforms on complex data. |
| **K-Nearest Neighbors** | ~0.005 s | Simple non-parametric design with zero training phase. | Computationally heavy inference phase; sensitive to feature scaling. |
| **Random Forest** | ~0.425 s | Robust handling of non-linear features; highly accurate. | Slower training overhead; reduced direct interpretability. |

## 5. Hyperparameter Tuning & Conclusion

**Bonus Optimization:** Advanced hyperparameter tuning was conducted using `GridSearchCV` on the Random Forest classifier over a defined parameter grid (`n_estimators`, `max_depth`, `min_samples_split`), optimizing directly for cross-validated F1 score.

**Final Recommendation:** The **Random Forest Classifier** is officially recommended as the optimal model for this task. Its ensemble architecture successfully captures intricate non-linear relationships and protects against overfitting, delivering superior evaluation metrics across all testing subsets.